

## 1. Data Overview

The dataset contains 77,536 historical loan applications, each described by seven borrower or loan characteristics together with a binary outcome variable indicating whether the loan was approved. The predictors include the requested loan amount, interest rate, borrower income, debt-to-income ratio, number of active accounts, number of derogatory marks, and total outstanding debt. The outcome variable, *loan\_status*, equals 1 for approved applications and 0 otherwise.

All variables are numeric and free from missing values, making the dataset suitable for statistical modelling without preprocessing. The approval rate is approximately 3.9%, indicating that the vast majority of applicants were not approved. This level of class imbalance is typical in banking environments where institutions adopt conservative credit policies. Overall, the dataset is large, clean, and well-structured, providing a solid foundation for developing a predictive screening model.

## 2. Model Choice and Rationale

The aim of this project is to build a screening model that estimates the likelihood that a new loan applicant will be approved. Given this objective and the binary nature of the outcome variable, logistic regression was selected as the primary modelling approach. Logistic regression is widely used in the financial industry—particularly in credit risk assessment—because it produces probability-based outputs that align naturally with lending decisions. For each applicant, the model generates an estimated probability of approval, allowing the bank to apply a clear and consistent decision threshold.

A major advantage of logistic regression is its interpretability. Unlike more complex machine-learning methods, logistic regression makes it possible to directly assess how each predictor affects approval likelihood. This transparency is important in regulated environments where lending decisions must be explainable to internal committees and external regulators. The method is computationally efficient, statistically robust, and supports standard diagnostic tools, such as hypothesis tests and goodness-of-fit measures. It also aligns with the quantitative techniques covered in the module. For these reasons, logistic regression offers a defensible and practical framework for developing a customer rating system for loan applications.

## 3. Model Development and Interpretation

To evaluate the model's predictive performance, the dataset was divided into an 80% training set and a 20% test set, with a fixed random seed to ensure reproducibility. The logistic regression model was initially fitted using all seven predictors. However, the variables *interest\_rate* and *total\_debt* were automatically excluded during model estimation due to perfect multicollinearity with other predictors. Their removal improves model stability without affecting predictive content, resulting in a final model based on five effective variables.

The regression coefficients highlight the characteristics most strongly associated with loan approval. Debt-to-income ratio is highly significant and negatively related to approval probability, indicating that applicants with high existing debt burdens are substantially less likely to be approved—a result consistent with standard credit-risk principles. Derogatory

marks are also significant and negatively related to approval outcomes, reflecting the importance of past credit behaviour in lending decisions. Other predictors, including loan size, borrower income, and number of accounts, were not statistically significant after accounting for the effects of the other variables. This suggests that within this dataset, their independent contribution to explaining approval decisions is limited, although they may still be relevant in broader underwriting practices.

Model fit statistics indicate that the predictors meaningfully improve explanatory power relative to a null model. The reduction in deviance and the model's AIC value demonstrate an acceptable balance between model complexity and predictive performance. While AIC could be compared with alternative modelling approaches for completeness, the logistic regression model already provides a stable and interpretable framework for estimating approval probabilities and supports its use as a practical screening tool.

#### 4. Model Evaluation

Confusion Matrix (Threshold = 0.0463)

| Predicted/Actual | 0     | 1   |
|------------------|-------|-----|
| 0                | 14780 | 470 |
| 1                | 111   | 147 |

Once fitted, the model's performance was assessed using the 20% test dataset. The model outputs a predicted probability of loan approval for each applicant. As expected in a highly imbalanced dataset, these probabilities are generally low, typically between 0.01 and 0.03. This reflects the fact that only 3.9% of applications in the historical data were approved.

Using the conventional threshold of 0.5, the model achieved an accuracy of 96.3%. However, because the vast majority of cases are "not approved," accuracy alone is not a meaningful indicator of performance. The confusion matrix reveals that the model correctly identifies 97% of the rejected applications (specificity = 0.969), but identifies only 57% of the approved applications (sensitivity = 0.570). Precision is relatively low at 0.238, reflecting the scarcity of positive cases in the dataset.

To provide a threshold-independent measure of discrimination, the ROC curve was constructed. The model achieves an area under the curve (AUC) of 0.7526, indicating good discriminatory power: in approximately 75% of randomly selected pairs consisting of one approved and one rejected application, the model assigns a higher predicted probability to the approved applicant. This level of performance is considered acceptable in credit-risk modelling and demonstrates that the model is capable of ranking applicants meaningfully according to creditworthiness.

## 5. Threshold selection

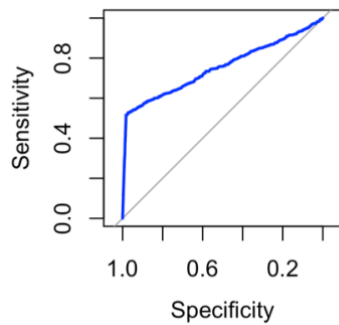


Figure 1. ROC Curve

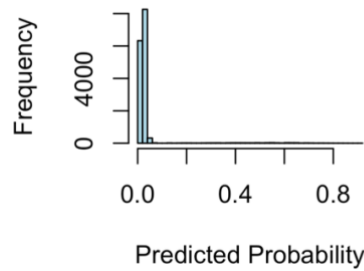


Figure 2. Distribution of Predicted Probabilities

Because logistic regression produces a continuous probability score, classification requires the choice of a threshold. The conventional threshold of 0.5 is not appropriate for this dataset, as no applicants achieve such a high predicted probability. Using 0.5 results in an overly conservative decision rule that classifies nearly all cases as “not approved.”

To identify a more suitable threshold, Youden’s Index was applied to the ROC curve. This method selects the threshold that maximises the sum of sensitivity and specificity, thereby providing the best overall balance between identifying creditworthy applicants and avoiding false approvals. The optimal threshold was found to be approximately 0.0463. At this value, the model achieves a sensitivity of 0.528 and a specificity of 0.972.

This threshold sits far below 0.5 due to the highly imbalanced nature of the dataset and reflects the bank’s historically cautious lending behaviour. Adopting a threshold within this lower range recognises that meaningful discrimination occurs among low predicted probabilities and provides a more practical and operationally aligned rule for classifying new applicants.

## 6. Recommendation for the bank

Based on the empirical results, the logistic regression model provides a reliable foundation for a loan-screening tool. It is highly effective at identifying applicants unlikely to be approved, which is central to managing credit risk. The model is also interpretable, enabling the bank to justify decisions and maintain transparency in regulatory environments.

However, because the dataset is heavily imbalanced, the model’s ability to identify approved applications remains moderate. For operational deployment, it is recommended that the bank adopt the ROC-derived threshold of 0.046 rather than the conventional 0.5, as this threshold provides a more realistic balance between approval volume and risk management.

The bank may choose to adjust this threshold depending on its risk appetite.

- A lower threshold would increase approval rates but also increase exposure to potential credit losses.
- A higher threshold would further reduce risk but at the cost of rejecting more potentially creditworthy borrowers.

Finally, although the logistic regression model performs well, the bank may consider exploring supplementary models—such as decision trees or random forests—in future work to determine whether predictive performance can be further improved without compromising interpretability.

## **7. Rating new applicants**

To demonstrate how the model can be applied operationally, a hypothetical new applicant was evaluated using the fitted logistic regression model. The applicant's profile included a loan size of €28,000, an income of €48,000, a debt-to-income ratio of 0.35, four active accounts, no derogatory marks, and total outstanding debt of €17,000. The model estimated an approval probability of approximately 3.3% for this applicant.

Using the ROC-derived threshold of 0.046, this applicant would be classified as “not approved.” This example illustrates how the model translates applicant characteristics into a quantitative approval probability and applies a consistent, threshold-based decision rule. The bank could use these predicted probabilities as internal risk scores to aid decision-making and to ensure uniform assessment across applications.

## **8. Conclusion from the chosen model**

This analysis developed and evaluated a logistic regression model to support the bank's loan-screening process using a dataset of 77,536 historical applications. The model is stable, interpretable, and effective at distinguishing between applicants likely and unlikely to be approved, as reflected in its AUC of 0.75. Debt-to-income ratio and derogatory marks emerged as the strongest predictors of approval, aligning closely with established credit-risk principles.

Given the dataset's strong class imbalance, the model produces relatively low approval probabilities, making the conventional 0.5 classification threshold inappropriate. The optimal threshold identified through ROC analysis (approximately 0.046) provides a more balanced and operationally meaningful decision rule. At this threshold, the model achieves high specificity and moderate sensitivity, consistent with a conservative lending environment.

Overall, the logistic regression model provides a robust and transparent basis for a customer rating system and can meaningfully support the bank's loan-screening process. With further development—such as exploring complementary models or incorporating additional applicant information—the predictive performance could potentially be enhanced while maintaining interpretability.

## **9. Limitations and future improvement**

While the logistic regression model provides a transparent and statistically grounded basis for screening loan applications, several limitations should be acknowledged. First, the dataset is highly imbalanced, with only 3.9% of applications approved. This imbalance constrains the model's ability to recognise approved applicants and contributes to relatively low sensitivity and precision. Although threshold optimisation helps address this, the underlying imbalance remains a structural challenge.

Second, the model relies solely on the variables provided in the dataset. Important factors that often influence lending decisions—such as employment history, credit utilisation ratios, length of credit history, or qualitative assessments—are not captured. Including a wider range of borrower characteristics could improve predictive performance and reduce the likelihood of borderline misclassifications.

Third, logistic regression assumes a linear relationship between the log-odds of approval and the predictors. Non-linear relationships or interaction effects may exist but are not explicitly modelled. More flexible modelling approaches, such as decision trees, random forests, gradient boosting methods, or regularised logistic regression (e.g., LASSO), could capture complex patterns more effectively while still allowing for interpretability.

Finally, the model was trained on historical data that reflects the bank's past lending behaviour. If lending policies or economic conditions change, the model may require periodic retraining to maintain performance. Ongoing monitoring, recalibration, and validation would therefore be essential for continuous operational use.

Despite these limitations, the model provides a strong foundation for a structured customer rating system. Enhancing the model with additional data, alternative algorithms, or regular performance monitoring could further improve its ability to support risk-based lending decisions.